

WSL installation

```
Command Prompt
Installing: Windows Subsystem for Linux
Windows Subsystem for Linux has been installed.
Installing: Ubuntu
Ubuntu has been installed.
The requested operation is successful. Changes will not be effective until the system is rebooted.

C:\Users\I _>wsl --version
WSL version: 2.1.5.0
Kernel version: 5.15.146.1-2
WSLg version: 1.0.60
MSRDC version: 1.2.5105
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.25131.1002-220531-1700.rs-onecore-base2-hyp
Windows version: 10.0.22631.3593

C:\Users\ _>
```

```
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: bosko
New password: 
Retype new password: 
passwd: password updated successfully
The operation completed successfully.
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.153.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
```

Commands

```
divya@Divyasree:~$ pwd
/home/divya
divya@Divyasree:~$ history
 1 sudo apt update
 2 sudo apt install build-essential
 3 gcc --version
 4 pwd
 5 date
 6 cal
 7 hostname
 8 history
 9 uname -a
10 whoami
11 pwd
12 ls
13 ls -l
14 ls -la
15 cd
16 date
17 gcc
18 jobs
19 exit
20 cal
21 date
22 pwd
23 mkdir OSSP
24 cd OSSP
25 pwd
26 mkdir src include obj bin docs tests scripts assets
27 ls
28 touch .gitignore README.md LICENSE Makefile
29 cd src
30 touch main.c shell.c parser.c executor.c builtin.c
```

```
102 history
divya@Divyasree:~$ whoami
divya
divya@Divyasree:~$ date
Mon Aug 3 12:12:57 UTC 2026
divya@Divyasree:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
divya@Divyasree:~$ sudo apt install ncal
[sudo] password for divya:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
ncal
0 upgraded, 1 newly installed, 0 to remove and 70 not upgraded.
Need to get 21.0 kB of archives.
After this operation, 59.4 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu/noble/universe amd64 ncal amd64 12.1.8 [21.0 kB]
Fetched 21.0 kB in 1s (30.4 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 46652 files and directories currently installed.)
Preparing to unpack ../archives/ncal_12.1.8_amd64.deb ...
Unpacking ncal (12.1.8) ...
Setting up ncal (12.1.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
divya@Divyasree:~$ whoami
divya
divya@Divyasree:~$ cal
August 2026
```

```
divya@Divyasree:~$ cal
August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
divya@Divyasree:~$ hostname
Divyasree
divya@Divyasree:~$ uname -a
Linux Divyasree 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
divya@Divyasree:~$ ls
OSSP practical2 process process.c
divya@Divyasree:~$ ls -l
total 28
drwxr-xr-x 11 divya divya 4096 Aug 1 08:10 OSSP
drwxr-xr-x  4 divya divya 4096 Aug 3 05:00 practical2
-rwxr-xr-x  1 divya divya 16312 Aug 3 03:53 process
-rw-r--r--  1 divya divya  775 Aug 3 03:58 process.c
divya@Divyasree:~$ ls -la
.      .bashrc      .motd_shown      practical2
..     .cache       .profile          process
.bash_history .config       .sudo_as_admin_successful process.c
.bash_logout .local        OSSP
divya@Divyasree:~$ cd
divya@Divyasree:~$ cd ossp
-bash: cd: ossp: No such file or directory
divya@Divyasree:~$ cd OSSP
divya@Divyasree:~/OSSP$ cd ..
divya@Divyasree:~$ mkdir divya
divya@Divyasree:~$ cd divya
```

```

divya@Divyasree:~/divya$ cd ..
divya@Divyasree:~$ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
divya@Divyasree:~$ find
.
./process.c
./practical2
./practical2/Task1
./practical2/Task1/command_executor
./practical2/Task1/command_executor.c
./practical2/Task2
./divya
./bashrc
./bash_history
./local
./local/share
./local/share/nano
./local/share/nano/search_history
./OSSP
./OSSP/LICENSE
./OSSP/tests
./OSSP/tests/test_parser.c
./OSSP/tests/test_executor.c
./OSSP/include
./OSSP/include/builtin.h
./OSSP/include/shell.h
./OSSP/include/executor.h
./OSSP/include/parser.h
./OSSP/README.md
./OSSP/docs
./OSSP/docs/design.md
./OSSP/docs/report.pdf

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./cache/mold.legal-displayed
divya@Divyasree:~$ locate
Command 'locate' not found, but can be installed with:
sudo apt install plocate
divya@Divyasree:~$ sudo apt install plocate
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  liburing2
The following NEW packages will be installed:
  liburing2 plocate
0 upgraded, 2 newly installed, 0 to remove and 70 not upgraded.
Need to get 160 kB of archives.
After this operation, 623 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 liburing2 amd64 2.5-1build1 [21.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 plocate amd64 1.1.19-2ubuntu2 [139 kB]
Fetched 160 kB in 2s (83.6 kB/s)
Selecting previously unselected package liburing2:amd64.
(Reading database ... 46660 files and directories currently installed.)
Preparing to unpack .../liburing2_2.5-1build1_amd64.deb ...
Unpacking liburing2:amd64 (2.5-1build1) ...
Selecting previously unselected package plocate.
Preparing to unpack .../plocate_1.1.19-2ubuntu2_amd64.deb ...
Unpacking plocate (1.1.19-2ubuntu2) ...
Setting up liburing2:amd64 (2.5-1build1) ...
Setting up plocate (1.1.19-2ubuntu2) ...
update-alternatives: using /usr/bin/plocate to provide /usr/bin/locate (locate) in auto mode
info: Selecting GID from range 100 to 999 ...
info: Adding group 'plocate' (GID 108) ...
Initializing plocate database; this may take some time...

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```

divya@Divyasree:~$ cd divya
divya@Divyasree:~/divya$ touch dsa ossp dbms
divya@Divyasree:~/divya$ tree
.
├── dbms
├── dsa
└── ossp

1 directory, 3 files
divya@Divyasree:~/divya$ cat

^C
divya@Divyasree:~/divya$ cat divya
cat: divya: No such file or directory
divya@Divyasree:~/divya$ cat ossp
divya@Divyasree:~/divya$ cp
cp: missing file operand
Try 'cp --help' for more information.
divya@Divyasree:~/divya$ mv dsa
mv: missing destination file operand after 'dsa'
Try 'mv --help' for more information.
divya@Divyasree:~/divya$ cd ..
divya@Divyasree:~$ mv dsa
mv: missing destination file operand after 'dsa'
Try 'mv --help' for more information.
divya@Divyasree:~$ mv dsa practical2
mv: cannot stat 'dsa': No such file or directory
divya@Divyasree:~$ rm dbms
rm: cannot remove 'dbms': No such file or directory
divya@Divyasree:~$ cd divya
divya@Divyasree:~/divya$ mv dsa practical2
divya@Divyasree:~/divya$ tree

```

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.
├── dbms
├── ossp
└── practical2

1 directory, 3 files
divya@Divyasree:~/divya$ rm ossp
divya@Divyasree:~/divya$ tree
.
├── dbms
└── practical2

1 directory, 2 files
divya@Divyasree:~/divya$ head
^C
divya@Divyasree:~/divya$ head dbms
divya@Divyasree:~/divya$ tail dbms
divya@Divyasree:~/divya$ more
more: bad usage
Try 'more --help' for more information.
divya@Divyasree:~/divya$ more practical2
divya@Divyasree:~/divya$ less
Missing filename ("less --help" for help)
divya@Divyasree:~/divya$ less ossp
ossp: No such file or directory
divya@Divyasree:~/divya$ less dsa
dsa: No such file or directory
divya@Divyasree:~/divya$ less dbms
divya@Divyasree:~/divya$ wc
^C
divya@Divyasree:~/divya$ wc dbms
0 0 0 dbms
divya@Divyasree:~/divya$ diff dbms practical2

```

```

divya@Divyasree:~/divya$ sort dbms
divya@Divyasree:~/divya$ uniq
^C
divya@Divyasree:~/divya$ uniq dbms\
> ^C
divya@Divyasree:~/divya$ uniq dbms
divya@Divyasree:~/divya$ grep dbms
^C
divya@Divyasree:~/divya$ cd ..
divya@Divyasree:~$ cd practical2
divya@Divyasree:~/practical2$ cd Task1
divya@Divyasree:~/practical2/Task1$ cd command_executor.c
-bash: cd: command_executor.c: Not a directory
divya@Divyasree:~/practical2/Task1$ nano command_executor.c
divya@Divyasree:~/practical2/Task1$ gcc command_executor.c -o command_executor
divya@Divyasree:~/practical2/Task1$ ./command_executor
Enter a Linux command: pwd

Parent Process
Parent PID : 1068
Child PID : 1069

Child Process
Parent PID : 1068
Child PID : 1069

Executing command: pwd
/home/divya/practical2/Task1

Child process completed.
divya@Divyasree:~/practical2/Task1$ which gcc
/usr/bin/gcc
divya@Divyasree:~/practical2/Task1$ whereis gcc

```

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gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz
divya@Divyasree:~/practical2/Task1$ man gcc
divya@Divyasree:~/practical2/Task1$ ps
  PID TTY          TIME CMD
   874 pts/4    00:00:00 bash
  1096 pts/4    00:00:00 ps
divya@Divyasree:~/practical2/Task1$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1        0  0 12:15 ?        00:00:00 /sbin/init
root         2        1  0 12:15 hvcd  00:00:00 /init
root         6        2  0 12:15 hvcd  00:00:00 plan9 --cont
root        52        1  0 12:15 ?        00:00:00 /usr/lib/sys
root       100        1  0 12:15 ?        00:00:01 /usr/lib/sys
systemd+   111        1  0 12:15 ?        00:00:00 /usr/lib/sys
systemd+   112        1  0 12:15 ?        00:00:00 /usr/lib/sys
root       172        1  0 12:15 ?        00:00:00 /usr/sbin/cr
message+   173        1  0 12:15 ?        00:00:00 @dbus-daemon
root       180        1  0 12:15 ?        00:00:00 /usr/lib/sys
root       182        1  0 12:15 ?        00:00:02 /usr/libexec
syslog     190        1  0 12:15 ?        00:00:00 /usr/sbin/rs
root       198        1  0 12:15 tty1    00:00:00 /sbin/agetty
root       211        1  0 12:15 ?        00:00:00 /usr/bin/pyt
root       300        2  0 12:15 pts/1   00:00:00 /bin/login -
divya      343        1  0 12:15 ?        00:00:00 /usr/lib/sys
divya      344      343  0 12:15 ?        00:00:00 (sd-pam)
divya      368      300  0 12:15 pts/1   00:00:00 -bash
polkitd    530        1  0 12:17 ?        00:00:00 /usr/lib/pol
root       866        2  0 12:42 ?        00:00:00 /init
root       868      866  0 12:42 ?        00:00:00 /init
divya      874      868  0 12:42 pts/4   00:00:00 -bash
divya     1097      874  0 13:02 pts/4   00:00:00 ps -ef
divya@Divyasree:~/practical2/Task1$ top
top - 13:02:34 up 47 min,  1 user,  load average: 0.10, 0.07, 0

```

```
divya@Divyasree:~/practical2/Task1$ top
top - 13:02:34 up 47 min, 1 user, load average: 0.10, 0.07, 0
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 5738.6 total, 5123.6 free, 537.8 used, 225.
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 5200.
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	21612	13188	9728	S	0.0	0.2
2	root	20	0	3180	2212	2072	S	0.0	0.0
6	root	20	0	3196	2132	2008	S	0.0	0.0
52	root	19	-1	66832	16112	14916	S	0.0	0.3
100	root	20	0	25416	6784	5128	S	0.0	0.1
111	systemd+	20	0	21344	13204	11084	S	0.0	0.2
112	systemd+	20	0	91036	7940	6968	S	0.0	0.1
172	root	20	0	4244	2684	2432	S	0.0	0.0
173	message+	20	0	9676	5420	4684	S	0.0	0.1
180	root	20	0	17980	8852	7812	S	0.0	0.2
182	root	20	0	1756104	11064	8632	S	0.0	0.2
190	syslog	20	0	222516	6032	4652	S	0.0	0.1
198	root	20	0	3124	1976	1836	S	0.0	0.0
211	root	20	0	107032	23172	13720	S	0.0	0.4
300	root	20	0	6700	4708	3928	S	0.0	0.1
343	divya	20	0	20320	11580	9452	S	0.0	0.2
344	divya	20	0	21164	3596	1868	S	0.0	0.1
368	divya	20	0	6080	5332	3664	S	0.0	0.1
530	polkitd	20	0	308172	8180	7280	S	0.0	0.1
866	root	20	0	3184	1124	980	S	0.0	0.0
868	root	20	0	3200	1260	1108	S	0.0	0.0
874	divya	20	0	6076	5420	3700	S	0.0	0.1
1098	divya	20	0	9284	5668	3500	R	0.0	0.1

```
divya@Divyasree:~/practical2/Task1$ kill 1234
-bash: kill: (1234) - No such process
divya@Divyasree:~/practical2/Task1$ kill -9 1234
-bash: kill: (1234) - No such process
divya@Divyasree:~/practical2/Task1$ jobs
divya@Divyasree:~/practical2/Task1$ bg %1
-bash: bg: %1: no such job
divya@Divyasree:~/practical2/Task1$ fg %1
-bash: fg: %1: no such job
divya@Divyasree:~/practical2/Task1$ ./hello &
[1] 1105
divya@Divyasree:~/practical2/Task1$ -bash: ./hello: No such file or directory
^C
[1]+  Exit 127 ./hello
divya@Divyasree:~/practical2/Task1$ chmod +x script.sh
Command 'chmod' not found, did you mean:
  command 'chmod' from deb coreutils (9.4-3ubuntu6.2)
Try: sudo apt install <deb name>
divya@Divyasree:~/practical2/Task1$ chmod 755 script.sh
chmod: cannot access 'script.sh': No such file or directory
divya@Divyasree:~/practical2/Task1$ ln file1 file2
ln: failed to access 'file1': No such file or directory
divya@Divyasree:~/practical2/Task1$ ln -s file 1 shortcut
ln: target 'shortcut': No such file or directory
divya@Divyasree:~/practical2/Task1$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0    2.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            2.9G  4.0K   2.9G   1% /mnt/wsl
drivers         237G  105G  133G  45% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.1G  954G   1% /
none            2.9G   36K   2.9G   1% /mnt/wslg
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none          2.9G   36K   2.9G    1% /mnt/wslg
none          2.9G    0   2.9G    0% /usr/lib/wsl/lib
rootfs        2.8G   2.8M   2.8G    1% /init
none          2.9G  520K   2.9G    1% /run
none          2.9G    0   2.9G    0% /run/lock
none          2.9G    0   2.9G    0% /run/shm
none          2.9G   80K   2.9G    1% /mnt/wslg/versions.txt
none          2.9G   80K   2.9G    1% /mnt/wslg/doc
C:\           237G  105G  133G   45% /mnt/c
tmpfs         574M   20K   574M    1% /run/user/1000
divya@Divyasree:~/practical2/Task1$ du -sh
20K
divya@Divyasree:~/practical2/Task1$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.6Gi       534Mi       5.0Gi       3.5Mi       253Mi       5.1Gi
Swap:          2.0Gi          0B       2.0Gi
divya@Divyasree:~/practical2/Task1$ id
uid=1000(divya) gid=1000(divya) groups=1000(divya),27(sudo),100(users)
divya@Divyasree:~/practical2/Task1$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=Divyasree
PWD=/home/divya/practical2/Task1
LOGNAME=divya
HOME=/home/divya
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/868_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31

```

```

XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/Git/cmd:/mnt/c/dev tools/apache-maven-3.9.12/bin:/mnt/c/Program Files/nod ej s:/mnt/c/Users/divya/.local/bin:/mnt/c/Users/divya/AppData/Local/Microsoft/WindowsApps:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Users/divya/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/divya/AppData/Roaming/npm:/mnt/c/Users/divya/AppData/Local/kin gsoft/WPS Office/12.1.0.27458/clitool:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
OLDPWD=/home/divya/practical2
_=usr/bin/env
divya@Divyasree:~/practical2/Task1$ cd export PATH=$PATH
-bash: cd: too many arguments
divya@Divyasree:~/practical2/Task1$ export PATH=$PATH
divya@Divyasree:~/practical2/Task1$ alias |='ls-l'
alias alert='notify-send --urgency=low -i "${[ $? = 0 ]} && echo terminal || echo error" "${history|tail -n1|sed -e '\''s/\s*[0-9]\+\s*//;s/[;|&]\s*alert$//'\`}'"
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
divya@Divyasree:~/practical2/Task1$ |

```